12TH BIOLOGY · CHAPTER 21 · SOLVED EXERCISE

# Immunity

Complete textbook exercise: MCQs, Short Questions, Long Questions and Inquisitive Questions

## SECTION 1: MULTIPLE CHOICE QUESTIONS

**1. Which of these are known for stimulating an immune response?**

(a) Pathogens (b) Antigens
(c) Antibodies (d) Histamines

**Answer: (b) Antigens** — Antigens are the foreign molecules that T and B cells recognize; they stimulate the production of antibodies.

**2. Which of the following is a part of the innate immune system?**

(a) T lymphocytes (b) B lymphocytes
(c) Skin and mucous membranes (d) Antibodies

**Answer: (c) Skin and mucous membranes** — Skin and mucous membranes are the first line of defence, which is present at birth. The others belong to acquired immunity.

**3. Which of these is not involved in body's non-specific defence?**

(a) Natural killer cells (b) Inflammatory response
(c) Antibodies (d) Complement system

**Answer: (c) Antibodies** — Antibodies are specific; they belong to the third line of defence.

**4. Skin offers an impenetrable barrier to microbes. It is categorized as;**

(a) Non-specific defence (b) Specific defence
(c) Acquired immunity (d) Passive Immunity

**Answer: (a) Non-specific defence** — Skin is the first line of defence; it acts against all microbes in the same way.

**5. What is the difference between B cells and T cells?**

(a) B cells make antibodies, and T cells destroy infected cells
(b) B cells and T cells do the same work
(c) B cells attack germs directly, and T cells make antibodies
(d) B cells fight germs in body fluids, and T cells work inside cells

**Answer: (a) B cells make antibodies, and T cells destroy infected cells** — B cells form plasma cells that secrete antibodies; cytotoxic T cells destroy infected body cells.

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**6. How do the neutrophils kill microbes?**

(a) By engulfing and digesting them (b) By producing antibodies
(c) By releasing hormones (d) By carrying oxygen to cells

**Answer: (a) By engulfing and digesting them** — Phagocytosis is the first method of neutrophils.

**7. Which group of chemicals is involved in inflammatory response?**

(a) Antibodies (b) Histamines and prostaglandins
(c) Prostaglandins and pyrogens (d) Antibodies and pyrogens

**Answer: (b) Histamines and prostaglandins** — Infected or injured cells release histamines and prostaglandins; pyrogens cause fever.

**8. Which type of white blood cell is responsible for producing antibodies?**

(a) Neutrophils (b) B lymphocytes (B cells)
(c) Macrophages (d) Natural killer cells

**Answer: (b) B lymphocytes (B cells)** — B cells divide into plasma cells, which secrete antibodies.

**9. What types of drug are not prescribed to a recipient of organ-transplantation?**

(a) Drugs that suppress immunity (b) Drugs that strengthen immunity
(c) Anti-inflammatory drugs (d) Anti-histamine drugs

**Answer: (b) Drugs that strengthen immunity** — A stronger immune system would attack the transplanted organ; the recipient is given drugs that suppress immunity.

**10. Cell-mediated immunity works best against;**

(a) Viruses infected cells and toxic molecules
(b) Viruses and toxic molecules
(c) Virus infected cells and cancer cells
(d) Toxic molecules and cancer cells

**Answer: (c) Virus infected cells and cancer cells** — Cytotoxic T cells kill virus-infected and abnormal (cancer) cells; toxic molecules are inactivated by antibodies.

## SECTION 2: SHORT QUESTIONS

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### 1. Define the roles of the different types of T cells.

| Kind of T cell | Role |
|---|---|
| Helper T cells | Initiate the immune response |
| Cytotoxic T cells | Break the cells that have been infected by microbes |
| Inducer T cells | Supervise the development of T cells in the thymus |
| Suppressor T cells | Terminate the immune response |

### 2. Skin is an important component of our defence system. Comment.

Skin is the first line of defence. Its outer cells form keratin, which makes it tough and impenetrable for microbes, and damaged cells of the stratum corneum are continuously shed and replaced. Its oil and sweat glands lower the surface pH to 3–5, which inhibits the growth of many microbes. Sweat also contains lysozyme, which digests the cell walls of many bacteria.

### 3. State how the digestive and respiratory tracts play a role in body's defence.

**Digestive tract:** Most bacteria present in food are killed by saliva in the oral cavity, which contains lysozyme. The strong digestive acids of the stomach and the protein-digesting enzymes of the intestine also kill bacteria.

**Respiratory tract:** Ciliated epithelial cells in the nasal cavity trap many bacteria of the air. The cells lining the smaller bronchi and bronchioles secrete sticky mucous that traps microbes before they reach the lungs. Cilia continually sweep this mucous upward, where it is swallowed.

### 4. How the natural killer cells play role in body's defence?

Natural killer cells do not kill the invading microbes. They detect and attack the body cells that have been infected with viruses. They release special proteins called **perforins**, which insert into the membrane of the target cell and form a pore. Water rushes in through the pore, and the cell swells and bursts. They also detect cancer cells and kill them before they can develop into a tumour.

### 5. Define pyrogens and state their effect of hypothalamus.

Pyrogens are chemicals released by macrophages when they attack invading microbes. They pass through the bloodstream to the hypothalamus, the body's thermostat. They stimulate it to raise the body's core temperature above the normal 37 °C, which results in fever (pyrexia).

### 6. What do you mean by autoimmune diseases?

Autoimmune diseases result when the immune system is unable to distinguish between body cells and foreign materials. So it attacks and destroys the cells and tissues of its own body. Examples are myasthenia gravis, systemic lupus erythematosis (SLE), rheumatoid arthritis and type I diabetes.

## 7. Differentiate between;

### • Antigen and antibody

| Antigen | Antibody |
|---|---|
| A foreign molecule, e.g., on an invading microbe. | A Y-shaped protein made by the body. |
| It stimulates the immune response. | It is secreted by plasma cells in response to an antigen. |
| It is marked for destruction. | It sticks to the specific antigen and marks it for destruction. |

### • B cells and T cells

| B cells | T cells |
|---|---|
| Develop and mature in the bone marrow. | Develop in the bone marrow but mature in the thymus. |
| Divide into plasma cells that secrete antibodies. | Do not make antibodies; four kinds: helper, cytotoxic, inducer and suppressor. |
| Key players of antibody-mediated immunity. | Carry out cell-mediated immunity. |

### • Macrophages and neutrophils

| Macrophages | Neutrophils |
|---|---|
| Develop from monocytes. | A type of white blood cell. |
| Ingest bacteria by endocytosis and kill them with lysozyme enzymes. | Kill by phagocytosis, by anti-microbial chemicals and by a network of DNA fibres. |
| Some are fixed in organs (lungs, liver, spleen, brain); they also release pyrogens and activate helper T cells. | They migrate from the blood to the inflamed tissue; dead neutrophils form part of pus. |

### • Complement system proteins and interferons

| Complement system proteins | Interferons |
|---|---|
| Produced in the liver; circulate in the blood in an inactive state. | Released by virus-infected cells. |
| Form a membrane-attack complex that punctures the plasma membrane of the microbe. | Make nearby cells enhance their anti-viral defences; activate NK cells and macrophages. |
| They kill microbes such as bacteria directly. | They stop viruses from infecting other cells. |

### • Innate and acquired immunity

| Innate immunity | Acquired immunity |
|---|---|
| Present at birth (inborn). | Acquired during growth. |
| Non-specific defence. | Specific defence against particular germs. |
| First and second lines of defence. | Third line of defence (immune system). |
| It has no memory. | It remembers previous invaders and responds quickly. |

### • Active and passive immunity

| Active immunity | Passive immunity |
|---|---|
| The body itself prepares antibodies against the antigens. | The individual is given antibodies. |
| It may be natural or artificial (vaccination). | Ready-made antibodies come from outside. |
| It is long-lasting. | It is short-lived. |

### • Natural active immunity and artificial active immunity

| Natural active immunity | Artificial active immunity |
|---|---|
| The body prepares antibodies naturally when exposed to antigens. | Weakened antigens are administered in the body (vaccination). |
| It follows a natural infection. | The weak antigens stimulate antibodies against the real antigens. |
| It is the most effective form of immunity. | It gives protection without suffering from the disease. |

### • Cell-mediated and Antibody-mediated immunity

| Cell-mediated immunity | Antibody-mediated immunity |
|---|---|
| Carried out by T cells. | Key players are B cells. |
| Cytotoxic T cells directly destroy the infected body cells. | B cells do not attack directly; antibodies mark antigens for destruction. |
| Works against virus-infected cells and cancer cells. | Works against antigens and toxic molecules; also called humoral immunity. |

## SECTION 3: LONG QUESTIONS

**1. Describe the role of macrophages and neutrophils in killing bacteria.**

Macrophages and neutrophils are germ-killer cells of the blood. They belong to the second line of defence (non-specific defence). They patrol the bloodstream and wait for invaders.

**Macrophages** develop from monocytes (a type of white blood cell). A macrophage ingests a bacterium by endocytosis and kills it with its lysozyme enzymes. Some macrophages are fixed in organs such as the lungs, liver, spleen and brain, but most of them patrol the blood, lymph and interstitial fluid.

**Neutrophils** are a type of white blood cell. They kill pathogens by three methods:

1. **Phagocytosis:** they engulf and digest the pathogens.
2. They release soluble **anti-microbial chemicals** on the pathogens.
3. They generate and release a **network of DNA fibres**, which bind with pathogenic bacteria and kill them.

During inflammation, more macrophages and neutrophils migrate from the blood to the infected tissue and engulf the microbes. Fever increases the speed of their action.

## 2. Explain how the proteins of the complement system kill bacteria and how the interferons inhibit the ability of viruses to infect cells.

The **complement system** consists of more than a dozen proteins made in the liver. They circulate in the blood in an inactive state. When they meet a microbe, they become active in a sequence: the first protein activates the second, and so on. The final five proteins form a **membrane-attack complex (MAC)** in the plasma membrane of the microbe. This complex punctures the membrane, so fluids and salts move into the microbe, and it swells and bursts.

**Interferons** are protective proteins released by virus-infected cells. They cause the nearby cells to enhance their anti-viral defences, so the virus cannot infect them easily. Interferons also activate other immune cells, such as natural killer cells and macrophages. They belong to the large class of proteins called **cytokines**, which work for communication between cells to trigger the protective defences of the immune system.

## 3. Inflammation is one of the most generalized nonspecific defences. How?

Inflammation is non-specific because it occurs in the same way against any infection or injury, whatever the microbe. Its events are:

1. Infected or injured cells release chemical signals such as **histamines** and **prostaglandins**.
2. These chemicals cause local expansion of blood vessels, so the capillary blood flow increases at the site. The capillaries also become more permeable.
3. This results in **redness, swelling and warmth** at the spot. The condition is called inflammation.
4. The heat makes the site unfavourable for microbial growth. It also promotes healing and raises the mobility of white blood cells.
5. More phagocytes (macrophages and neutrophils) migrate from the blood to the inflamed tissue and engulf the microbes.
6. Monocytes arrive and release chemicals that kill the microbes; then macrophages clean up dead microbes, cells and debris.

The inflammatory response is often strong enough to stop the spread of microbes.

## 4. How are monocytes, T-cells and B-cells the components of the immune system?

The immune system is not localized in one place. It is composed of individual white blood cells that arise in the bone marrow, circulate in blood and lymph, and rush to the site of infection. They are also found in the lymph nodes, spleen, liver, thymus and bone marrow. There are three main kinds: monocytes, T cells and B cells (T and B cells are together called lymphocytes).

### 1. Monocytes

**Monocytes** are a type of WBC. They circulate in the blood and enter the interstitial fluid, where they mature into **macrophages**. Macrophages phagocytose and destroy microbes, and produce proteins that activate helper T cells. Monocytes also mature into other phagocytes that stay in the liver, spleen, lymph nodes and other tissues. They help to clear cellular debris and repair tissue damage.

### 2. T cells

**T cells** develop from stem cells in the bone marrow and migrate to the **thymus** (a small grey gland just above the heart). There they mature and gain the ability to identify the antigens of invading microbes. Tens of millions of different T cells are made, each specialized for one particular antigen. There are four main kinds:

| Kind of T cell | Role |
|---|---|
| Helper T cells | Initiate the immune response |
| Cytotoxic T cells | Break the cells that have been infected by microbes |
| Inducer T cells | Supervise the development of T cells in the thymus |
| Suppressor T cells | Terminate the immune response |

### 3. B cells

**B cells** also develop in the bone marrow, but they do not travel to the thymus; they complete their maturation in the bone marrow and are then released into the blood and lymph. Each B cell is specialized to recognize a particular antigen. When a B cell meets its antigen, it divides into many **plasma cells**. Each plasma cell produces an antibody that sticks to the antigen and marks it for destruction.

## 5. Differentiate the two types of acquired immunity (active and passive immunity).

| Active immunity | Passive immunity |
|---|---|
| The body itself prepares antibodies against the antigens of microbes. | The individual is given antibodies to combat specific microbes. |
| It may be natural (exposure to antigens) or artificial (vaccination). | Ready-made antibodies come from outside the body. |
| Memory cells are formed, so it is long-lasting. | It is short-lived. |

Active immunity is of two kinds. In **natural active immunity**, the body prepares antibodies naturally when exposed to antigens; it is the most effective form of immunity. **Artificial active immunity** is achieved by administering weakened antigens (vaccination), which stimulate the body to prepare antibodies against the real antigens.

## 6. Describe the roles T-cells in cell-mediated immunity.

When a macrophage engulfs a microbe, it produces special proteins for the helper T cells. The helper T cells recognize these proteins through their receptors and become activated. The activated helper T cells start two parallel immune responses: cell-mediated immunity and antibody-mediated immunity.

Cell-mediated immunity is carried out by T cells. It protects the body from virus infections and cancer by killing abnormal or virus-infected cells. Its events are:

1. **Proliferation:** The activated helper T cell produces soluble chemicals, e.g., T cell growth factor. It starts the proliferation of all types of T cells, so large clones of T cells that recognize the antigen are formed.
2. **Activation:** A second factor from the helper T cell attracts macrophages to the site of infection.
3. **Induction:** Helper T cells activate inducer T cells in the thymus, which trigger the maturation of lymphocytes into T cells.
4. **Attack:** Receptors on the cytotoxic T cells recognize the infected body cells. The cytotoxic T cells bind with them and disrupt their membranes, so the cells burst.
5. **Suppression:** Suppressor T cells multiply more slowly than cytotoxic T cells. After 1 to 2 weeks their number rises enough to shut down the cytotoxic T cell response.
6. **Memory:** A population of helper and cytotoxic T cells persists, probably for life, as **memory cells**. They give a rapid response if the same microbe enters again.

## 7. Describe the role of B-cells in antibody-mediated immunity.

Antibody-mediated (humoral) immunity is a second, stronger defence activated by the helper T cells. Its key players are the B cells. They do not attack antigens directly; they mark them for destruction and also inactivate toxic molecules. Its events are:

1. **Proliferation:** Each B cell binds to the antigen through the B receptor proteins on its surface. Helper T cells recognize such antigen-bound B cells, bind with them and release proteins that stimulate the B cell to proliferate. A large number of **plasma cells** are produced.
2. **Differentiation and secretion:** Most plasma cells stop reproducing and start making more B receptor proteins. When secreted, these proteins are called **antibodies**. A great number of antibodies are secreted in only a few days.
3. **Attack:** Antibodies mark the antigen for destruction in three ways. (i) *Complement:* complement proteins make holes in the antibody-coated antigen cell, which swells and bursts. (ii) *Macrophages:* they recognize the antibody-coated antigen, ingest it and break it. (iii) *Killer cells:* they have receptors for antibody-coated antigens; they bind and kill the cell.

4. **Suppression:** After several weeks, suppressor T cells shut down the antibody response.
5. **Memory:** Some plasma cells do not produce antibodies; they persist as **memory B cells**. They give a rapid secondary response if the same antigen enters again.

## 8. Define allergies. Correlate the symptoms of allergies with the release of histamines.

**Allergies** result from the hypersensitivity of the immune system to weak antigens that do not cause an immune response in most people. The substances that cause allergies are called **allergens**, e.g., dust, molds, pollen, cat dander, certain foods and some medicines (such as penicillin).

### Histamines and the symptoms

1. After exposure to an allergen, some people make a kind of antibody called **E-antibodies**.
2. These antibodies bind to **mast cells** (found in connective tissue around blood vessels).
3. The mast cells release **histamines**, which start the inflammatory response.
4. Histamines cause dilation of blood vessels and leakage of fluid. This produces the symptoms: sneezing, runny nose, itching etc.
5. In some individuals, histamines cause life-threatening **anaphylactic shock**, in which swelling makes breathing difficult.

**Treatment:** Antihistamine drugs block histamines and give temporary relief. In **desensitization**, extracts of the antigens are injected into the patient. This produces G-antibodies in the blood, which deal with the allergen before it meets the E-antibodies.

## 9. How are T-cells and B-cells related to transplant rejections?

Normally, T and B cells attack only foreign molecules and cells, because every person's cells carry a particular set of **self-proteins**, like molecular "fingerprints", which the person's own T and B cells recognize.

When a person receives an organ transplant, the self-proteins of the donated organ may not match those of the recipient. Then the recipient's **T cells attack the cells** of the transplanted organ and the **B cells produce antibodies** against its proteins. This is called **transplant rejection**.

### How rejection is minimized

- Doctors look for a donor whose self-proteins match the recipient's as closely as possible. Preference order: identical twin > sibling > parent > unrelated donor.
- Drugs are used to suppress the recipient's immune system; but they also reduce the ability to fight infections.
- Monoclonal antibodies are used to destroy the T cells that attack the transplant.
- Stem cells are used to make a new immune system that recognizes the transplant as self.

## INQUISITIVE QUESTIONS

### 1. Rationalize the inflammatory response in arthritis as an example of a misdirected immune response.

Inflammation is meant to act against infection or injury. In arthritis, inflammation occurs in the joints although there is no infection or injury there. In rheumatoid arthritis, the patient forms antibodies against the cartilaginous tissue of his own joints. The immune system treats this body tissue as foreign, so the joints become inflamed (swollen, warm, painful) and are damaged. The response is directed at the body itself instead of at microbes, so it is a misdirected immune response.

### 2. Justify why the physicians prescribe antipyretic drugs, when fever is a nonspecific defence against microbial infections.

Mild fever is useful: it hinders the reproduction of microbes and speeds up neutrophils and macrophages. But high fever is harmful to the body itself. Temperatures above 103 °F (39.4 °C) are considered dangerous, and those above 105 °F (40.5 °C) are often fatal. So physicians give antipyretic drugs to keep the temperature within safe limits and to relieve the patient, while the other defences and medicines act against the microbes.

### 3. Justify why physicians prescribe antihistamine therapy to the patients of runny nose or skin rashes.

Runny nose, sneezing, itching and rashes are symptoms of allergy. In allergy, E-antibodies bind to mast cells, which release histamines. Histamines dilate the blood vessels and make fluid leak out, which produces these symptoms. Antihistamine drugs block the histamines, so the symptoms are reduced and the patient gets temporary relief.

### 4. Describe the discovery of monoclonal antibodies and justify how this accomplishment revolutionized many aspects of biological research.

Only a small portion of an antigen, called the **determinant**, fits into the binding site of an antibody. Antibodies that attach to different portions of the same antigen are **polyclonal antibodies**. Antibodies that attach with only one determinant are **monoclonal antibodies**.

**Discovery:** Cesar Milstein and George Kohler received the Nobel Prize in 1984 for learning how to produce a monoclonal antibody. They mixed plasma cells (which produce antibodies) with cancer cells called **myelomas**. New cells called **hybridomas** grew in the culture. Hybridomas used the genes of the plasma cell to make the antibody and the myeloma genes to grow and divide. So they obtained rapidly growing cells that kept producing one monoclonal antibody.

#### How it revolutionized biological research

- They are used to purify specific molecules from complex mixtures; interferon was purified in this way.
- In medicine, they are used as vehicles for delivering specific therapies.

- Research is under way to target radioactive monoclonal antibodies against antigens found only on cancer cells, to kill cancer cells selectively.
- They are used to destroy the T cells that attack a transplant, to prevent transplant rejection.

Before this discovery, a pure antibody against one determinant could not be obtained in large amounts. Hybridomas give an unlimited supply of one identical antibody, which is a very exact tool for finding, purifying and targeting one specific molecule.

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